

1. Environment

a person steps in
front of the wall

2. Sensors

limited range; noise:
one wall 2 m away
reads 1.92 to 2.07 m

robot
where am I?

3. Robot (actuation)

wheels slip, gears wear:
“move 1 m” is
not exactly 1 m

4. Models

the kinematic model
ignores slip and bumps

5. Computation

answers must come in time,
so algorithms approximate